

Multimedia Databases Exercises SS 2026

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Exercise 8

Subject: Querying
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Task 1: Difference between relational and object-relational databases

Outline the main differences between relational and object-relational database systems. Explain in particular the complexity of the used data structures, their indexing, the request predicates and the advantages and disadvantages of each type of system.

Task 2: DB Definitions

Explain the following notions related to object-relational databases in your own words. Give examples for each notion (more information on this topic at: Oracle Docs):

- User-defined data type
- Inheritance
- Table of objects
- Polymorphism
- OID
- Relations between user-defined data types (1:1, 1:n)
- REF and Deref
- Collections
- Dot-operator

Task 3: Content-based Image Search Using SQL/MM

The goal of this exercise is to develop a DB-application to manage a zoo. The DB must include information about **animals**, **keepers** and **cages**. **Animals** are characterized by a *Name*, *Gender*, *Type*, *Age* and an *image*. **Keepers** are characterized by a *Name* and a *Pass-photo*. **Cages** are assigned a *pitch*, and at least 1 *keeper*. The application offers the possibility to search for **animals**, based on SQL/MM features.

1. Sketch the entity-relationship model (ER model) for the given DB.
2. Write object-relational SQL statements to create the DB
3. Write SQL/MM queries to content-based search for animals (Query-by-Example)

Task 4: MPEG Query Format (MPQF)

Formulate the following natural language queries, into the corresponding MPQF queries. Consider the MPEG-7 standard data model as a starting point.

1. Search for images and their titles, which are similar to a presented example image, taken in Munich, and not bigger than 2048K. Take the similarity criterion as the most important.
2. Search for images, which have in their northern part a mountain and their southern part a lake. Furthermore, the images must obey the following dominant color distribution (Red 10%, Green 30% and Blue 60%). The search must return only JPEG images, and 20 at the maximum. Results must sorted by the title, similarity reference, and the file size.